

EDUCATION

Stony Brook University

Stony Brook, NY

- B.S. Candidate in Computer Science | Sophomore Standing
- Honors College | Honors in Computer Science Program

May 2030

Stuyvesant High School

New York, NY

- Excellence in Mathematics Award
- Clubs: Stuyvesant Math Team, Mu Alpha Theta Mathematics Honor Society, Spectator / Indicator Photos
- Relevant Coursework: AP Calculus BC, Linear Algebra, Cybersecurity, AP Physics C (Mech & E/M)

GPA: 97.64/100

EXPERIENCE

Memorial Sloan Kettering Cancer Center

New York, NY

Student Intern, Medical Physics

May 2025 - November 2025

- Fine-tuned and validated a CNN-based deformable image registration (DIR) model for aligning cone-beam CT scans to planning CTs for lung cancer treatment monitoring, achieving validation DICE score over 0.75
- Experimented with a multi-head attention architecture for DIR, reaching parity with CNN and VoxelMorph baselines
- Wrote MATLAB and Python preprocessing and testing code used for preparing training datasets
- Ran training and inference on HPC clusters and configured CUDA environments for training compatibility

Summer Intern

July 2024 - August 2024

- Wrote C# and Python preprocessing code for head/neck cancer CT segmentation training for treatment tracking
- Leveraged HPC clusters and IBM LSF (bsub) for job scheduling

Hack Club

Shelburne, VT and Remote

Hardware Engineering Intern

May 2022 - March 2024

- Designed and prototyped production PCBs for Blot (2D CNC drawing machine) and Sprig (RP2040-based hand-held gaming console), including motor control and USB-C PD boards; Sprig reached a 600+ unit production run
- Published open-source reference designs adopted by the maker community (NXP/NTAG I2C NFC board, USB hub with USB-C implementation guide) and mentored dozens of high schoolers through first-time PCB design as part of the Onboard program

Logistics and Operations Intern

July 2021 - August 2023

- Built a mailing address database using USPS Intelligent Mail Barcode (IMb), enabling no-cost trackable postage; reduced domestic shipping costs up to 60% and international costs up to 80%, including devising alternate routing through regional trade disruptions
- Integrated the Shippo API into a Slack-based shipping agent to automate label generation and cost comparison
- Supported operations for in-person hackathons in San Francisco (approx. 150 attendees), Vermont, and New York

PROJECTS

Personal Site and Photography Portfolio | Next.js, Tailwind CSS, Cloudinary CDN | [GitHub](#) 2026

- Designed a performant static site with EXIF metadata rendering and flexible category / collection filtering
- Implemented an authenticated admin dashboard (NextAuth, Bcrypt) with Cloudinary CDN asset delivery and a Supabase PostgreSQL backend via custom API wrappers

Component Inventory Manager | Next.js, PostgreSQL (Supabase), NextAuth | [GitHub](#) 2026

- Built a full-stack inventory tracking system to manage electronic components, MPNs, and distributor data, and generate asset ID labels
- Built API wrappers to query, create, and update components by MPN, description, and category

AWARDS & HONORS

- AIME Qualifier (3x)
- CEMC Euclid/Fermat School Winner (2025)
- Gold President's Volunteer Service Award

SKILLS

Languages: C, C++, Python, MATLAB, C#, Java

Frameworks: Next.js, HTML/CSS, Supabase

Tools: Git, Linux, KiCAD, LaTeX